

MRFE: A Deep-Learning-Based Multidimensional Radio Frequency Fingerprinting Enhancement Approach for IoT Device Identification

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Abstract—Nowadays, wireless networks have been widely deployed in our daily lives, providing people with convenient Internet of Things (IoT) services in healthcare, smart cities, transportation, etc. However, the open nature of communication mediums leaves IoT devices susceptible to unauthorized access by rogue devices, leading to significant privacy breaches and property damage. Among various security measures, radio frequency (RF) fingerprinting stands out as a promising device identification technique, owing to RF fingerprints' uniqueness and forgery-resistant nature. Existing methods, however, overlook the structural relationship of a transmitter's internal hardware paths, affecting the performance and efficiency of RF fingerprint identification. Inspired by the internal hardware paths, this article introduces a novel deep-learning-based RF fingerprinting approach, multidimensional RF fingerprinting enhancement (MRFE). MRFE enhances RF fingerprinting by dissecting raw IQ signals into multiple dimensions and proposing a novel fingerprint strengthen layer (FSL) to extract multidimensional fingerprints from the separate hardware paths, then leveraging attention mechanisms to fuse them into an enhanced RF fingerprint. The enhanced fingerprint captures more detailed physical hardware characteristics, effectively enhancing device identification accuracy. Our MRFE's open-source implementation has been validated on the public ORACLE RF fingerprinting data set, achieving an impressive 99.33% accuracy in identifying 16 high-end bit-similar transmitters with identical configurations.

Index Terms—Deep learning, Internet of Things (IoT) device identification, IoT security, IQ signal, radio frequency (RF) fingerprint identification, RF fingerprinting (RFF), wireless network security.

I. INTRODUCTION

WIRELESS networks, extensively used in fields like communication, transportation, and healthcare, offer significant conveniences to our daily lives. By 2025 [1], the number of global wireless radio frequency (RF) connections is expected to soar to approximately nine billion, which notably

includes connections for Internet of Things (IoT) devices. However, the inherent open nature of wireless communication mediums makes IoT devices susceptible to security breaches. These breaches are often perpetrated by rogue IoT devices that can mimic legitimate device identities to infiltrate private networks [2], execute malicious operations, and access confidential data [3], thereby leading to considerable privacy leakage and property damage [4], [5]. In this context, device identification becomes a critical defense mechanism. It aims to accurately recognize and authenticate the identity of devices within a network, which is paramount for effective network resource management and monitoring. By ensuring reliable device identification, organizations can safeguard sensitive information, preserve the integrity of their networks, and mitigate potential risks associated with unauthorized access.

RF fingerprinting (RFF) is a promising physical layer device identification approach that relies on the device intrinsic hardware impairments of RF components [6]. Specifically, manufacturing limitations lead to slight variances in oscillators, power amplifiers (PAs), and analog filters, even among devices of the same brand, model and production line. These variances, intrinsic to the hardware, create unique physical-layer fingerprints for each device, thus making it challenging to forge [3]. In addition, RF fingerprinting has the advantages of low cost and flexibility, as it does not necessitate device modification or cooperation, facilitating easy deployment and integration with other security measures.

Among existing RF device identification methods, deep-learning-based approaches outperform artificial feature-based methods [7]. The former utilizes the powerful learning ability of the neural network to automatically extract RF fingerprints in the physical signal, which avoids complex feature engineering. They also provide a comprehensive analysis of multiple hardware impacts at the transmitter [8]. In contrast, artificial feature-based methods focus only on single hardware influences, often underestimating the impact of other components on the signal. Meanwhile, due to the limited feature space of the artificially designed fingerprint, it can only identify a small number of devices, rendering them less practical for large-scale applications [9], [10]. Deep-learning-based RF fingerprinting, free from these constraints, offers a more robust solution for device identification.

With the explosive growth number of IoT devices, there is an urgent need to develop a neural network that could extract advanced RF fingerprints based on various hardware

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information at the transmitter [11]. In deep-learning-based methods, four neural network models are widely utilized, i.e., convolutional neural networks (CNNs), generative adversarial networks (GANs), probabilistic neural networks (PNNs), and autoencoders [12]. However, the GAN is difficult to train and optimize, since it involves a complex adversarial process between two neural networks that compete with each other. PNNs require a large number of neurons and memory space to approximate the unknown density function, especially if the data is high-dimensional or complex. CNNs may make RF fingerprints less efficient or accurate, since the important or useful features may not be properly extracted or analyzed [13]. Autoencoders face challenges, such as the potential overfitting, reliance on the architecture and input data quality, as well as high computational demands. Although they are adept at feature learning and dimensionality reduction [4], [14], they inherently lack classification capabilities, necessitating additional mechanisms for effective application in RFF tasks.

Adjusting the neural network to align with RF signal characteristics is essential for obtaining advanced RF fingerprints. In the transmitter, the hardware paths for generating the I (in-phase) and Q (quadrature) signals are independent of each other. These paths comprise two sets of functionally identical hardware components and the two individual signals contain some unique hardware fingerprints, such as IQ imbalance, phase noise, and carrier frequency offset. However, our empirical study reveals that the RF signals input to neural networks in current research are simply mixtures of I and Q signals [15]. Consequently, this makes it challenging for the neural network to separate them and extract their own features accurately. Furthermore, most methods overlook the structural relationship of the hardware path transmitting IQ signals, significantly diminishing RF fingerprinting's practicability and speed.

In this article, we propose a novel RF fingerprint enhancement and identification method named multidimensional RF fingerprinting enhancement (MRFE). Inspired by the internal hardware paths, MRFE enhances RF fingerprinting by dissecting raw IQ signals into multiple dimensions and proposing a novel fingerprint strengthen layer (FSL) to extract multidimensional fingerprints (MulDi-RF fingerprints) from the separate hardware paths, then leveraging attention mechanisms to fuse them into an enhanced RF fingerprint. The enhanced fingerprint captures more detailed physical hardware characteristics, thereby boosting device identification accuracy. To evaluate the proposed MRFE, we have conducted extensive experiments based on publicly available data sets. The experimental results demonstrate that the advantages of MRFE are primarily manifested in two aspects: 1) enhanced accuracy and 2) efficient learning. MRFE achieves an impressive 99.33% accuracy in device identification, even in the most challenging scenario where the transmitters are a large pool of bit-similar devices (same hardware, protocol, physical address, and MAC ID).

The main contributions of this article are summarized as follows.

- 1) We propose an RF fingerprint enhancement method for wireless device identification named MRFE. Compared

with the existing methods, our enhanced RF fingerprint captures more detailed physical hardware characteristics, enabling more efficient feature learning and effectively boosting device identification accuracy.

- 2) We have designed a neural network plugin, FSL, which constructs three independent branches for assessing feature weights across various dimensions, thus emphasizing crucial information in the MulDi-RF fingerprint.
- 3) The implementation of MRFE and five other advanced models has been extensively tested using the publicly available ORACLE RF Fingerprinting data set [16], which consists of raw IQ samples collected from 16 high-end bit-similar X310 USRPs. The results demonstrate that MRFE achieves the highest accuracy at 99.33%, outperforming all other tested models.
- 4) We have made the source code of MRFE publicly available and it can be accessed for free via <https://github.com/YZKQDU/MRFE>.

The remainder of this article is structured as follows. In Section II, related work is introduced. Then, the proposed MRFE detection technique is detailed in Section IV. In Section V, we evaluate the proposed method through extensive experiments. Finally, we conclude this article in Section VI.

II. RELATED WORK

This section categorizes existing wireless device identification methods into two groups based on the RF fingerprint extraction process: 1) artificial feature-based schemes and 2) deep-learning-based schemes. We summarize the approaches focusing on RFF tasks in Table I, comparing them across various dimensions.

A. Artificial Feature-Based Identification Schemes

Artificial feature-based identification schemes differentiate wireless devices using deliberately designed features as RF fingerprints. Designing artificial RF fingerprints requires a rich knowledge base, such as the fundamental principles of wireless communication, the hardware structure of wireless devices, and the status variations in transmission channels. Existing artificial feature methods mainly employ modulation domains, statistical features and transient variation to generate RF fingerprints. The modulation domain-based schemes use signal characteristics like IQ imbalance, carrier frequency, and spectrum analysis to create RF fingerprints. Ding et al. [17] proposed a method to extract RF fingerprints based on the IQ imbalance of orthogonal signals. IQ imbalance is caused by hardware differences in the orthogonal modulator, which is unique and can be used to identify the transmitter. Nguyen et al. [18] found that the carrier frequency can be used to construct a unified wireless device identification scheme, which can be applied to Bluetooth, Lightbridge, Wi-Fi, XPD, and ZigBee devices. Jafar et al. [19] focused on creating robust RF fingerprints via spectrum analysis, suitable for a wide signal-to-noise ratio range. However, the modulation domain features of RF signals are seriously affected by interference signals and multipath effects, leading to changes and distortions.

TABLE I
COMPARISON WITH PRIOR WORKS IN RFF TASKS

Related works	Framework	Input data	AFE	PDS	Emitter type	FGI	Performance
Ding et al. [17]	SVM	I/Q imbalance data			12 phones; 6 USRPs	✓	$Acc = 85.34\%$
Nguyen et al. [18]	SYL-4	RF images			6 devices of 5 types		$Pre = 99.14\%$
Li et al. [21]	CNN	Density trace figure			5 transmitters		$Acc \geq 99\%$
Zhou et al. [24]	XGBoost	Instantaneous characteristics		✓	3 same emitters	✓	$Acc = 96.4\%$
Zhuang et al. [25]	SVM	51 statistical RF features			6 base stations	✓	$Pre \geq 99\%$
Yang et al. [26]	SRC	3 RF features			8 similar walkie-talkies	✓	$Acc = 98.75\%$
Reising et al. [27]	SVM	RF-DNA fingerprints			18 WiMAX radios	✓	$Acc \geq 90\%$
Kose et al. [28]	PNN	Spectral fingerprints			8 different wifi devices		$Acc = 97.9\%$
Taşcıoğlu et al. [29]	PNN	Amplitude + PCA features			8 wifi devices		$Acc = 97.50\%$
Han et al. [30]	GAN	I/Q imbalance data	✓		4 emitters		$Acc = 97.36\%$
Zhang et al. [31]	DANN	Raw I/Q signals + ideal modulation signals	✓		7 distinct emitters		$Acc \approx 98\%$
Yang et al. [32]	GAN	Raw I/Q signals			10 Hack RFs		$Acc = 98.10\%$
Feng et al. [33]	CNN-GRU	Raw I/Q signals	✓	✓	16 high-end bit-similar USRPs	✓	$Acc = 98.70\%$
Bremnes et al. [34]	CNN	Sub-band signals		✓	15 UAVs		$Acc = 97.25\%$
Li et al. [35]	CNN	Gray image			4 distinct mobile phones		$Acc \geq 90\%$
Chavez et al. [36]	CNN/MLPNN	Dual-layer waveform images			12 Zigbee devices	✓	$Acc = 98.54\%$
Sankhe et al. [37]	CNN	Raw I/Q signal	✓	✓	16 high-end bit-similar USRPs	✓	$Acc = 98.6\%$
This work	MRFE	I/Q + I + Q signals	✓	✓	16 high-end bit-similar USRPs	✓	$Acc = 99.33\%$ $Pre = 99.32\%$

Notes: AFE stands for Automatic Feature Engineering; PDS is an abbreviation for Public Dataset; FGI refers to Fine Grained Identification.

Statistical feature-based identification schemes utilize various statistical characteristics of RF signals to generate device fingerprints, such as weighted centroid, density trajectory and cumulative information. Khandker et al. [20] proposed using the weighted centroid of multiple RF signals for device identification. Li et al. [21] extracted density trajectory as an RF fingerprint from distorted constellation diagrams. This trajectory represents the nonlinear effects of PAs, which can uniquely identify hardware devices. Shao and Chen [22] used the cumulative information of IQ data to generate RF fingerprints. The feature curve formed by this cumulative information can comprehensively analyze the subtle characteristics of wireless devices. Ali and Fischer [23] formed RF fingerprints by statistically analyzing transmitted signal's amplitude and phase errors. Phase errors, arising from unique nonlinear hardware impairments in transmitters, are challenging to imitate. However, statistical feature-based schemes are significantly impacted by channel noise and require substantial data.

Transient variation-based identification schemes extract RF fingerprints from signal transients. The transient refers to a period before the actual transmission of data, which contains information such as amplitude fluctuations and phase differences of electromagnetic waves. Zhou et al. [24] employed gradient boosting of transient frequency, phase, and amplitude for RF fingerprints, comprehensively considering various transient changes. To avoid the impact of dynamic outdoor environments on signals, Zhuang et al. [25] attempted to construct RF fingerprints using 51 statistical features in modulation domain and waveform domain. This fingerprint can be used for fake base station discovery, location and tracking. Yang et al. [26] employed sparse representation-based classification (SRC) to classify the proposed RF fingerprints, which consist of rising transient, falling transient, and sync features of the signal. Reising et al. [27] applied the discrete Gabor transform to generate the transient features of RF signals. This method extracts more RF features by analyzing signals in both

time and frequency domains. However, transient variation-based identification schemes require precise starting points and transient duration, otherwise, the accuracy of device identification will be severely reduced.

B. Deep-Learning-Based Identification Schemes

Deep-learning-based identification schemes leverage neural networks to extract and identify unique fingerprints from wireless signals. The network can directly process raw signals without complex transformations. Additionally, these networks automatically learn abstract signal features, thereby reducing the need for extensive feature selection engineering. This scheme can utilize various neural networks, such as PNNs, GANs, and CNNs. The PNN is a model based on statistical principles. It uses summation layers to weigh and sum the signal, obtaining the classification probability for each device. Köse et al. [28] employed a PNN to classify the RF fingerprints of WiFi transmitters, which could reduce the complexity of signal transformation. Taşcıoğlu et al. [29] combined principal component analysis (PCA) with PNNs for wireless device identification. PCA is used to reduce signal dimensionality, obtaining important features, while PNN is used to perform feature classification. However, the PNN-based identification scheme is susceptible to outliers, leading to poor stability.

GAN-based identification schemes consist of a generator and a discriminator. The generator in this network automatically learns from wireless device transmitters, thereby simulating the real RF signals. Meanwhile, the discriminator is used to distinguish real and generated RF signals, improving identification accuracy and robustness. Lalouani et al. [38] employed an adversarial network to detect the protocol configuration of signal transmitters, which is applicable to both encrypted and open transmissions. Han et al. [30] proposed a neural network consisting of multiple discriminators. Each discriminator is trained in an adversarial framework and acts

as a classifier to distinguish RF fingerprints. Zhang et al. [31] demodulate and reconstruct IQ signals by domain-adversarial neural networks to generate invariant fingerprint features. Yang and Yan [32] inputted noise into the generator to generate a fake signal distribution. The discriminator is trained with the real and fake signal distributions. The feedback from the discriminator is used to refine the generator. Over multiple updates, the discriminator can distinguish legitimate senders from illegitimate ones. However, GANs are limited in generating a large variety of samples, restricting their suitability for identifying numerous devices.

CNN-based identification schemes utilize convolution kernels to extract RF fingerprints. The kernel uses signal-to-noise ratio or other channel information to select the appropriate region, which could alleviate signal instability caused by sleep mode switching and temperature changes. Feng et al. [33] proposed combining a convolutional network with a gated recurrent unit for RF fingerprint identification. This method employs convolutional layers to extract the orthogonal and correlated features of IQ signals and uses gated recurrent units to extract temporal features. The above two mechanisms can reduce the training parameters while ensuring identification accuracy. Bremnes et al. [34] applied convolutional networks to unmanned aerial vehicle recognition. The vehicle RF signal is decomposed into 16 sub-band signals with fixed boundaries, which are fed into a lightweight convolutional network to quickly identify devices. Li et al. [35] converted the RF signal into a 2-D representation through image processing, and then use a classic CNN to identify the fingerprint grayscale image. Chavez et al. [36] generated dual-layer waveform domain images from the raw signals and combined them with a multilayer perceptron neural network and convolutional network to identify RF fingerprints. Sankhe et al. [37] introduced ORACLE, a CNN-based model that achieves 98.6% accuracy using raw I/Q samples with 16 high-end, bit-similar devices in static conditions. Despite the widespread use of CNNs in RF fingerprinting, the uniformity of convolution kernel weights can obscure important features, affecting identification time and accuracy.

III. BACKGROUND KNOWLEDGE

The RF fingerprint can be extracted from the RF signal for device identification. An RF fingerprint primarily refers to the unique characteristics of RF signals that can be attributed to the specific hardware of a transmitter. This technique exploits the minor imperfections and variations inherent in the manufacturing process of RF components, which cause each transmitter to exhibit slightly different signal characteristics, even among devices of the same make and model. The process of RF generation is illustrated in Fig. 1, which is implemented by multiple hardware, including oscillators, filters, PAs, etc.

The process begins with digital signal processing (DSP), where the original data (voice, video, text, etc.) is encoded, modulated and prepared for transmission. Importantly, this is where the data is split into two separate streams to create the I and Q components. The DSP-processed I and Q digital signals are then converted into their analog counterparts using

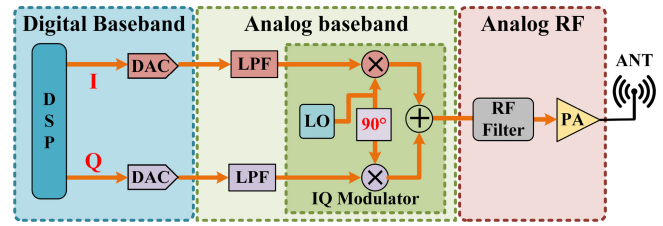


Fig. 1. Process of generating RF signals.

two digital-to-analog converters (DACs), one for each signal. This conversion is essential for the signals to be modulated onto a carrier wave for RF transmission. The analog I and Q signals may pass through low-pass filters (LPFs) to remove high-frequency artifacts from the DAC process, ensuring the signals remain within the intended bandwidth and reducing spectral leakage. The core of the RF transmission process, the I/Q modulator, takes the analog I and Q signals and modulates them onto a carrier wave generated by a local oscillator (LO) [39]. This modulation shifts the baseband signals to the RF domain, effectively combining them into a single composite RF signal. The I signal modulates the carrier directly while the Q signal modulates a 90° phase-shifted version of the carrier, enabling the encoding of data in both the amplitude and phase of the RF wave. After modulation, the RF signal is passed through an RF filter and a PA to ensure it can cover the required distance [40], [41]. Finally, the amplified RF signal is transmitted into the environment through an antenna (ANT), which radiates the energy as electromagnetic waves.

As clearly illustrated in Fig. 1, the in-phase (I) signal and the quadrature (Q) signal are typically generated through two parallel signal processing paths. These paths utilize distinct DACs, LPFs, and mixers, thereby embedding unique hardware fingerprints into the I and Q signals, which can provide additional information for device identification. However, current methods for extracting RF fingerprints tend to treat the I and Q signals as a unified entity, neglecting the structural relationship of the hardware paths. This oversight significantly diminishes the efficacy and speed of RF fingerprinting in existing RF-based device identification methods.

IV. PROPOSED MECHANISM

In this section, we propose MRFE. The framework of MRFE and the structure of the proposed FSL are described in detail.

A. Overview of MRFE

MRFE can achieve precise and fine-grained RF Fingerprinting using only raw IQ data at the physical layer. As shown in Fig. 2, MRFE consists of three parts, i.e., the signal processing module, the feature extraction module, and the fingerprint fusion module.

Signal Processing Module: This module is responsible for processing the raw IQ signal into multidimensional RF signals (I signal, Q signal, and IQ signal) using the innovative random selection value (RSV) method. This method can introduce randomness into the signal band, effectively simulating real-world RF signal acquisition scenarios.

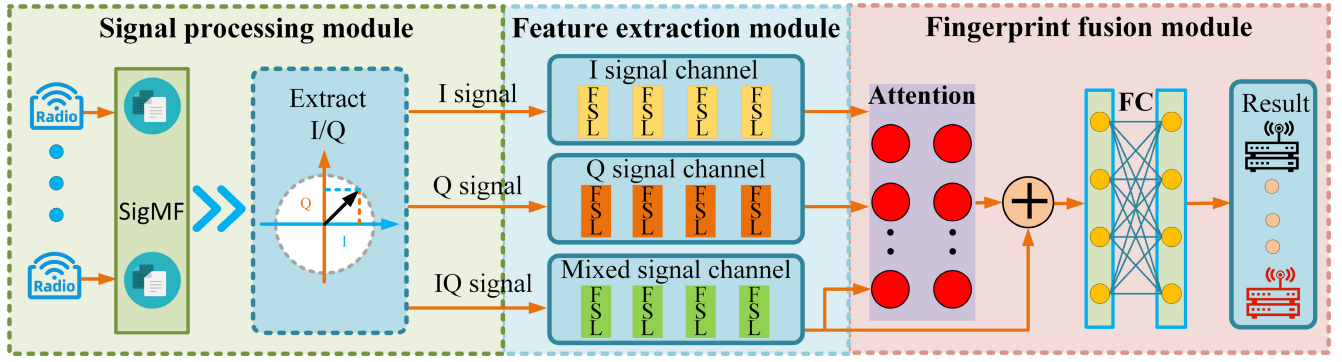


Fig. 2. Overall framework of MRFE.

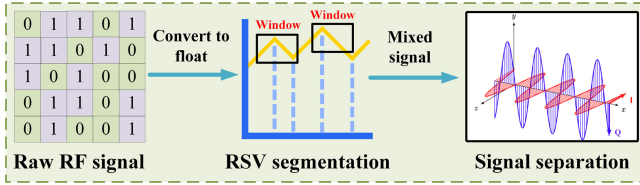


Fig. 3. Process of the signal processing module.

Feature Extraction Module: This module is responsible for extracting MulDi-RF fingerprints from the multidimensional RF signals. As depicted in Fig. 2, this module is composed of three distinct signal channels, and each channel consists of four FSLs for specific MulDi-RF fingerprint extraction and enhancement. This module outputs MulDi-RF fingerprints: the I signal fingerprint, Q signal fingerprint, and the mixed-signal fingerprint.

Fingerprint Fusion Module: This module is responsible for fusing multidimensional fingerprints extracted from feature extraction module to generate the final enhanced RF device fingerprint. Such an enhanced RF fingerprint exhibits enhanced feature representation ability, which can be used for accurate device identification purpose.

B. Signal Processing Module

This module processes the raw IQ signal into multidimensional RF signals for neural network input. This is a necessary step because the multidimensional RF signals contain more intrinsic hardware information and are more suitable for the proposed model. Overall, the signal processing process can be divided into three steps, as depicted in Fig. 3. First, converting binary raw IQ data collected by the RF receiver into float format. Second, the RSV method determines the segment input for the feature extraction module. Finally, the data within the segment is separated into I signals, Q signals, and IQ signals.

Specifically, for the second step, we use the RSV method to segment the float file and randomly select IQ signal segments for further signal separation. That is, each segment has a random initial position, so real-world RF signal collection processes can be simulated by this method. The RSV method includes two functions: data segmentation and random selection. The process of data segmentation involves partitioning

the float IQ signals into segment with uniform lengths through manipulation of the window size and sliding stride parameters. This operation expands the scale of the training set, which facilitates network training. The random selection process is employed to assign a random index to each segment. This operation has two advantages.

- 1) The RF collection is inherently random in applications. Random selection is more congruent with actual circumstances.
- 2) Different information is contained in the various RF signal. Randomly selecting signals helps the network model to extract hardware fingerprints, which are independent of content.

For the third step, we separate the IQ signal segment into multidimensional RF signals for feeding the proposed neural network. The signal contained in each segment is stored in a complex form. The real part represents the I signal, while the imaginary part is the Q signal. To fully utilize the RF signal, we extract the I and Q signals separately. This operation divides the raw RF signal into three categories: 1) I signal; 2) Q signal; and 3) IQ signal. The aforementioned procedure offers dual benefits, i.e., amplification of the data volume and augmentation of the information available.

C. Feature Extraction Module

The feature extraction module aims to extract the MulDi-RF fingerprints, i.e., I signal fingerprint, Q signal fingerprint, and mixed-signal fingerprint. As mentioned in Section III, because I and Q signals are generated through two independent signal processing paths, they can encapsulate distinct hardware features and provide more information beyond mixed IQ signals. So, in this module, we use such three-path signals as input data and innovatively design a three-channel model to explore the unique features in each type of multidimensional RF signal (I signal, Q signal, and IQ signal). Such three-channel model includes an I signal channel, a Q signal channel, and a mixed-signal channel, which consists of our innovative FSL plugin.

In the model, to extract the MulDi-RF fingerprints, three independent channels are applied, i.e., the I signal channel, the Q signal channel, and the mixed-signal channel. The I signal channel exclusively processes the I signal, the Q

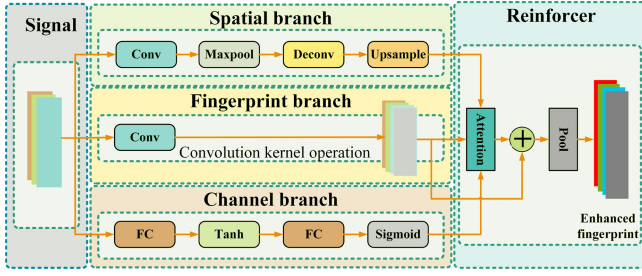


Fig. 4. Structure of FSL.

signal channel solely handles the Q signal, and the mixed-signal channel is dedicated to processing the mixed IQ signal. As shown in Fig. 2, the three channels operate in parallel to prevent interference among hardware fingerprints. Each channel comprises four innovative FSLs. The serial connection of the four FSLs ensures that the neural network has sufficient learning capabilities to record hardware fingerprints. The I signal channel outputs the I signal fingerprint, and the Q signal channel outputs the Q signal fingerprint. The mixed-signal channel outputs the mixed-signal fingerprint. The I signal and Q signal fingerprints, respectively, denote the corresponding hardware features from distinct signal processing paths. The mixed-signal fingerprint encapsulates all pertinent hardware characteristics of the transmitter. The details of FSL are presented in Section IV-E.

D. Fingerprint Fusion Module

This module fuses multidimensional fingerprints to produce the final enhanced RF device fingerprint. The enhanced fingerprint exhibits enhanced feature representation ability, which can be used for device identification.

The fingerprint fusion module consists of an attention layer, a residual connection, and two fully connected layers. The attention mechanism highlights key positions in the mixed-signal fingerprint, drawing on the I signal fingerprint and Q signal fingerprint. This operation can increase the information contained in the mixed fingerprint. The calculation formula for the attention layer is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

where Q represents the output of the Q signal channel, K is the output of the I signal channel, and V represents the output of the mixed-signal channel. T represents the transpose operation, and d_k denotes the input signal length.

This module introduces a novel device identification method using residual connections to fuse outputs from the mixed-signal channel and the attention layer. This fusion creates an enhanced fingerprint that encompasses all relevant hardware information and can accelerate the network's training speed. The final stage of identification involves reducing the fingerprint's dimensionality through a fully connected layer.

E. Structure of FSL

FSL is a plugin we proposed, which can widely apply in deep learning to measure the feature weight from different dimensions.

In this article, we use FSL to highlight the crucial elements of the RF fingerprint. FSL consists of four components: 1) the fingerprint branch; 2) the channel branch; 3) the spatial branch; and 4) the reinforcer, as depicted in Fig. 4. This section outlines the structure and principles of FSL.

The fingerprint branch extracts the fingerprint of input signals, the channel branch generates the weight of each fingerprint, and the spatial branch generates the weight of each part in the fingerprint. The reinforcer highlights the corresponding part of the fingerprint based on the channel weight and spatial weight.

1) *Fingerprint Branch*: The fingerprint branch is used to extract the fingerprint from input signals. It consists of a convolutional layer that contains multiple kernels. The specified stride controls the kernels' movement, which perform element-wise multiplication with subregions of the input signal to generate feature values. After iterating through each subregion of the input signal, a fingerprint for the input signal can be obtained. In deep learning models, each kernel automatically updates to extract varied hardware fingerprints of the signal. The calculation formula of the convolutional layer is

$$\text{out}(N_i, C_j) = \text{bias}(C_j) + \sum_{k=0}^{C_n-1} W(C_j, k) \star \text{input}(N_i, k) \quad (2)$$

where $\star$ denotes an effective cross-correlation operation. N_i represents the batch size of training data, C_j is the number of channels, and k represents the length of the signal sequence.

Empirical studies have shown that in previous approaches, each convolution kernel in a convolutional network has the same weight, resulting in the same weight for each fingerprint. However, in practical applications, the weight of each fingerprint is different, and the weight of each part in the fingerprint should also be different. Therefore, we introduce the channel branch to determine each fingerprint's weight and the spatial branch to assess the weight of each fingerprint part.

2) *Channel Branch*: The channel branch is designed to generate the weight of each fingerprint. It consists of two fully connected layers and two activation functions. The first fully connected layer reduces the data's dimensionality within the channel, converting the RF fingerprint into 1-D data. The Tanh activation function performs a nonlinear transformation on the data. Furthermore, the data transformation range of the Tanh function is $[-1, 1]$, which is zero-centered. This data distribution is advantageous in neural network training to circumvent the vanishing gradient problem. The second fully connected layer transforms the number of channels in the branch to match the number of channels in the fingerprint branch. The second activation function, the sigmoid function, produces outputs ranging from $[0, 1]$, utilized for calculating the weights of each channel. The calculation formula of the fully connected layer is

$$y = xA^T + b \quad (3)$$

where x is the input signal, A denotes the weight matrix, T represents the transpose operation, and b is the bias.

3) *Spatial Branch*: The spatial branch is constructed to generate the weight of each fingerprint part. It consists of a

convolutional layer, a max-pooling layer, a deconvolutional layer, and an upsampling layer. The convolutional layer conducts a convolution operation on a specific channel and performs data reduction by channel dimensionality reduction. This operation serves to minimize any potential spatial weight distortions of the fingerprint by effectively eliminating channel interference. The max-pooling layer selects the maximum value as the output in each subregion of the input tensor to enhance key features. The deconvolutional layer restores the input data size of the convolutional layer, with the output emphasizing the salient features of the fingerprint. The upsampling layer reshapes the data to match the fingerprint branch's shape. The sigmoid function calculates the weight of each part of the fingerprint. The calculation formula for the max-pooling layer is

$$L_{\text{out}} = \left\lfloor \frac{L_{\text{in}} + 2P - D(K - 1) - 1}{S} + 1 \right\rfloor \quad (4)$$

where L_{out} is the output of the max-pooling layer, and L_{in} denotes the input of the max-pooling layer. P represents the filling value, S is the step size of the sliding window. K represents the size of the sliding window, and D denotes the expansion value of the sliding window.

4) *Reinforcer*: The reinforcer highlights the corresponding part of the fingerprint based on the channel weight and spatial weight. It consists of an attention layer and a pooling layer. In the attention layer, the Spatial and channel branches function as Q and K , respectively, with the fingerprints from the fingerprint branch serving as V . The attention layer emphasizes crucial fingerprint parts using spatial and channel weights, enhancing the model's accuracy. The fingerprint branch's output is linked to the attention layer through a residual connection, which avoids the vanishing and exploding gradient problems. The subsequent pooling layer reduces the fingerprint dimensionality to reduce the parameter number and network computational complexity. These methods help prevent overfitting and enhance the network's generalization capabilities. The calculation formula of the reinforcer is

$$R_{\text{out}} = \text{pool}(V + \text{attention}(Q, K, V)) \quad (5)$$

where Q , K , and V represent the output of the spatial branch, the channel branch, and the fingerprint branch, respectively.

V. EVALUATION

In this section, we first describe the data set details, performance metrics, and experimental parameters. Then, we compares MRFE with current state-of-the-art methods in wireless device identification. Finally, the ablation experiments using ORACLE data set are conducted.

A. Experimental Setup

1) *Hardware and Software Setup*: The experiment in our study are performed on a DELL OptiPlex 5090 PC, equipped with an Intel i5-3320M CPU. The laptop operates on a Linux 4.2.0 kernel and runs the Ubuntu 14.04 LTS operating system, providing a stable and robust environment for our experiments.

2) *Data Set Details*: To evaluate the MRFE, we employ the publicly available real-world ORACLE RF Fingerprinting data set [16]. The data set is collected in an open area with much less reflections. It includes raw IQ signal samples collected from 16 high-end, bit-similar X310 USRP SDRs, with the same B210 radio as the receiver [37]. Notably, these transmitters are uniform in hardware and present a narrower range of impairments, which considerably limits the scope for differentiation. Sharing the same components, composition, protocol, physical address, and MAC ID, they transmit data frames over the air conforming to the IEEE 802.11a standard with random payloads. This setup provides a stringent test of device identification, relying solely on the ability to distinguish RF hardware fingerprints.

Additionally, we use a synthetic data set (Data Augmentation RF Fingerprinting Data set-Dataset#2) [42], [43] to preliminarily evaluate the performance of MRFE and five other models under different SNR conditions, especially at low SNR levels. This data set was generated using the MATLAB Communications Toolbox to simulate ten virtual radios, with SNR levels ranging from -10 to 20 dB by adjusting the additive white Gaussian noise.

In both data sets, we randomly select 80% of the data as the training set, 10% as the validation set, and the remaining 10% as the testing set.

3) *Performance Metrics*: For a comprehensive evaluation of MRFE's device identification capability, four metrics are utilized: 1) accuracy; 2) precision; 3) recall; and 4) F1-score. Accuracy is the ratio of correctly identified RF devices to the total number of devices. This metric assesses MRFE's overall equipment identification ability. Precision is the ratio of correctly identified RF devices to those predicted as belonging to this category. Recall measures the ratio of correctly identified RF devices to the total devices in the same category. Precision and recall measure MRFE's device identification capability for each category. The F1-score, combining precision and recall, measures MRFE's overall identification capability.

4) *Experimental Parameters*: In the signal processing module, the window size is 512, and the sliding step is 256, ensuring adequate signal information. In the feature extraction module, four FSLs are used, bolstering MRFE's learning capability. In the fingerprint fusion module, the number of fully connected layers is 2 to ensure the accuracy of RF fingerprint mapping. The training batch size is set to 64, considering training speed, memory usage and stability. The initial learning rate is set at 0.002 and halved every 20 epochs to prevent overfitting. MRFE is implemented in PyTorch and trained on a DELL OptiPlex 5090 PC. Its source code is publicly available can be accessed via <https://github.com/YZKQDU/MRFE>.

B. Comparison With State of the Arts

To validate the superiority of the MRFE, we study five other state-of-the-art models in wireless device identification field and conduct comparative experiments with MRFE.

1) *AlexNet-1D* [44]: The AlexNet-1D neural network is an improved version of the AlexNet neural network [45]

TABLE II
PERFORMANCE COMPARISON BETWEEN MRFE AND
STATE-OF-THE-ART MODELS

Model	Parameter count	Layer count	Mean accuracy (%)	Convergence epoch	Inference time
AlexNet	654752	10	95.64±0.05	17±1	0.083s
ARFNet	547892	15	97.55±0.01	13±1	0.072s
CVCNN	985421	13	98.13±0.10	15±2	0.134s
ResNet	743823	10	96.26±0.06	16±1	0.089s
Inception	1143823	9	97.03±0.04	15±1	0.158s
MRFE	871046	14	99.33±0.01	9±1	0.093s

designed to handle 1-D data from RF signals. The AlexNet-1D distinguishes itself from traditional CNNs with a deeper network structure and a greater number of convolutional kernels. The AlexNet-1D also uses ReLu activation functions and Dropout regularization methods to avoid overfitting.

2) *ARFNet* [46]: The ARFNet neural network leverages IQ signal information for wireless device identification. It consists of multiple DAConv layers, pooling layers, and fully connected layers. The DAConv layers automatically extract fingerprints from RF signals. These fingerprints are then converted into device identification results through fully connected layers and a softmax activation function.

3) *CVCNN* [47]: The CVCNN neural network is a type of complex-valued neural network. Instead of converting IQ signals to floats, it inputs the neural network directly in a complex form. CVCNN alters the weight computation within convolutional layers by incorporating complex values, enabling it to process complex-valued IQ signals.

4) *ResNet* [48]: ResNet is a type of CNN. It features residual blocks with skip connections that allow the input from one layer to be added to the output of a layer deeper in the network. This architecture enables the training of networks that are much deeper than was previously feasible, significantly improving performance on tasks requiring recognizing complex patterns.

5) *Inception* [49]: The Inception network, also known as GoogLeNet, employs “Inception modules” that consist of parallel convolutions with varied filter sizes, pooling operations, and 1×1 convolutions for dimensionality reduction, enabling efficient learning of complex features. This design enhances the network’s depth and breadth without sacrificing computational efficiency, allowing it to capture information across multiple scales effectively.

To compare the state-of-the-art models with our MRFE, we train and test the above models on the ORACLE data set [16]. We present the performance metrics of our models including accuracy, convergence epochs, parameter count, layer count, and inference time in Table II. The MRFE model, consisting of 871 046 weights across 14 layers, achieves the highest accuracy at 99.33%, outperforming all other models tested. This superior performance is further illustrated in Fig. 5, which shows MRFE’s training dynamics alongside five other models. MRFE distinguishes itself by reaching convergence

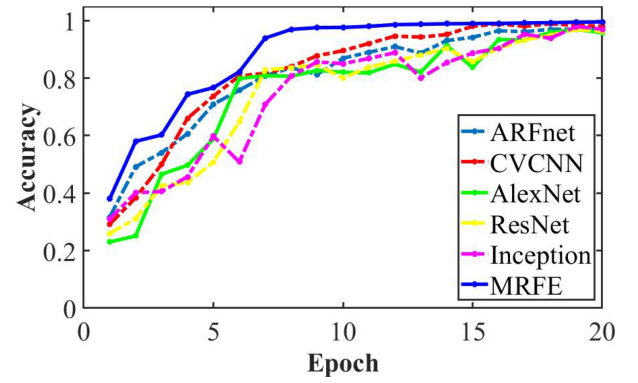


Fig. 5. Training process of MRFE and state-of-the-art models.

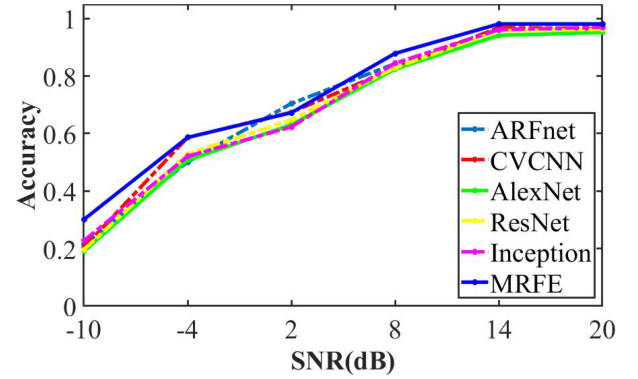


Fig. 6. Results of the six models under different SNR conditions (Synthetic data set).

in just 9 epochs, demonstrating both efficiency and effectiveness in learning. Additionally, it maintains stable accuracy beyond epoch 9 with only minor fluctuations, indicating robust performance compared to models like AlexNet and Inception, which exhibit greater variability in accuracy. This stability, coupled with its leading accuracy, highlights MRFE’s effectiveness in RF fingerprinting for device identification. Furthermore, MRFE demonstrates an inference time of 0.093 s and an average operating power of 45 W, showcasing its efficiency in real-time applications.

To further compare the performance of the six models under different SNR conditions, especially at low SNR levels, we have conducted experiments using the Data Augmentation RF Fingerprinting Data set-Dataset#2. Fig. 6 illustrates the performance of the six models on this synthetic data set. The results indicate that the accuracy of MRFE can surpass 98.2% when the SNR exceeds 14 dB. However, as the SNR decreases from 8 to −10 dB, the accuracy rates of all models exhibit a significant decline. This suggests that extracting hardware features from the signal becomes exceedingly difficult as noise obscures or distorts the crucial attributes of the raw IQ signals, complicating the precise extraction of essential features.

C. Property Analysis

Based on the evaluation results, we present a property analysis of MRFE focus on three aspects.

Accuracy: The MFRE model's ability to achieve a 99.33% accuracy rate on a challenging data set of 16 bit-similar RF transmitters with identical configurations underscores its exceptional precision in differentiating between devices with highly similar characteristics, reflecting its reliability even under demanding circumstances.

Robustness: In evaluation experiments conducted using the ORACLE data set, the robustness of MRFE is particularly exemplified by its training on data sets collected in a laboratory setting, which incorporates random environmental noise. Moreover, the varying distances between transmitters and receivers, extending from 0.6 to 18.9 m at 1.8 m intervals, further stress test the model under realistic and fluctuating conditions. The model's ability to maintain high accuracy across such diverse scenarios demonstrates its resilience and robust handling of real-world complexities.

In evaluation experiments conducted using the synthetic data set, MRFE could surpass 98.2% accuracy rate when the SNR exceeds 14 dB, which further demonstrates that MRFE can maintain robustness under moderate SNR conditions.

Scalability: Under the experimental setup described in Section V-A, MRFE can complete its training in merely 10 min and execute inference operations within 0.1 s, illustrating its remarkable efficiency. Such expeditious training and inference rates are pivotal for the model's scalability, facilitating its quick deployment and enabling continuous iterative enhancements. These features are particularly beneficial in scenarios where rapid response and prudent resource management are demanded, thereby underscoring MRFE's substantial scalability and adaptability in diverse operational contexts.

D. Effect of RSV

RSV is an essential component in the signal processing module. The method randomly selects the data inputted into the MRFE to help the neural network extract RF fingerprints independent of signal content. In this section, we conduct RSV ablation experiments on MRFE and five other state-of-the-art models with the same experimental settings as the previous section. Note, all subsequent ablation experiments were conducted using the ORACLE data set. During model training, the input data is categorized into two types: DS-RSV and DS-NoRSV. DS-NoRSV consists of the original and unprocessed signal data, which will be input into the neural network in order; DS-RSV represents the signal data processed by RSV, which will be input into the network in random order by RSV. The experiment employs four metrics: accuracy, precision, recall, and F1-score, to comprehensively evaluate the performance of the above models. The result is shown in Table III.

The experimental results presented in Table III demonstrate that, irrespective of whether data sets processed with RSV (DS-RSV) or unprocessed data sets (DS-NoRSV) are used, the MRFE consistently outperforms all other models. This consistent superiority highlights the effectiveness of our proposed method. Furthermore, it is notable that all models experience approximately a 4.6% average improvement in performance when utilizing DS-RSV, as compared to their performance

TABLE III
RESULTS OF RSV ABLATION EXPERIMENTS

Input data	Model	Accuracy	Precision	Recall	F1 score
DS-NoRSV	AlexNet	91.54%	91.62%	91.51%	91.56%
	ARFNet	93.41%	93.42%	93.42%	93.42%
	CVCNN	93.10%	93.41%	93.02%	93.22%
	ResNet	91.48%	91.69%	91.32%	91.50%
	Inception	92.34%	92.46%	92.01%	92.38%
	MRFE	94.26%	94.01%	94.33%	94.16%
DS-RSV	AlexNet	95.64%	95.75%	95.62%	95.68%
	ARFNet	97.55%	97.53%	97.57%	97.55%
	CVCNN	98.13%	98.11%	98.15%	98.13%
	ResNet	96.26%	96.38%	96.13%	96.25%
	Inception	97.03%	96.73%	97.32%	97.02%
	MRFE	99.33%	99.32%	99.35%	99.33%

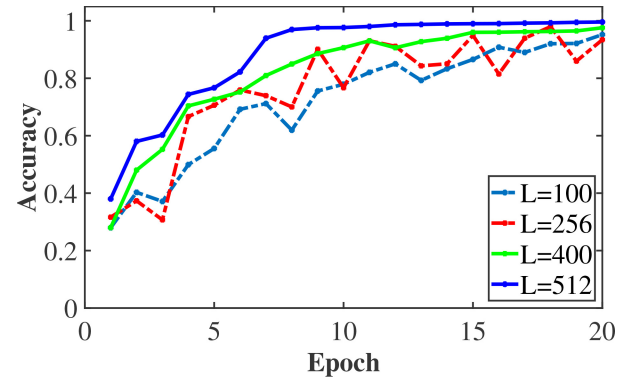


Fig. 7. Training process with different input signal lengths.

with DS-NoRSV. These enhancements are evident across all metrics, including accuracy, precision, recall, and F1 score, confirming that RSV processing can effectively enhance device identification capabilities in both the MRFE and other state-of-the-art models.

E. Effect of Input Signal Length

In the signal processing module, RF signals are segmented into the same lengths to suit neural network training requirements. Signal length, a critical hyperparameter, needs to be determined based on experiments. If the length is too short, the carried RF information is insufficient for wireless device identification. Conversely, if the length is too long, it will increase the time of data collection and the burden the collection equipment. This section investigates the optimal signal length for device identification in MRFE by varying the input signal length.

The model is trained with different input signal lengths. In the experiment, L represents the number of floats in each input signal segment. The segment length is divided into four cases: 100, 256, 400, and 512. Fig. 7 reveals that with an input data length of 100, the model has its lowest accuracy at 90%. Moreover, the convergence speed of this length is the slowest. At an input data length of 256, the model's convergence speed accelerates, yet its accuracy fluctuates significantly. At an input

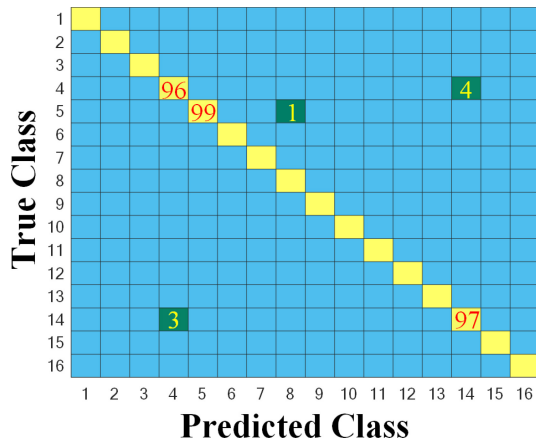


Fig. 8. Illustration of MRFE's confusion matrix for 16 bit-similar devices.

data length of 400, although the model's accuracy becomes stable, it still has significant improvement space. With an input length of 512, MRFE achieves an optimal accuracy of 99.33%. Additionally, the model is stable and the convergence speed is fast, which can meet the RF equipment detection requirements. While the input data length can be further extended, it would considerably increase RF signal collection time and model detection time. So, considering various factors, the optimal input signal length is determined to be 512.

F. Effect of RF Fingerprint

Identifying devices of the same brand and model is the most challenging device identification task. The data set described in Section V-A is used to construct such scenario where all device identification factors are maintained consistency except for the RF fingerprint. This enable MRFE to identify devices solely based on their RF hardware fingerprints.

The confusion matrix for the 16 experimental devices, displayed in Fig. 8, shows an overall accuracy of 99.33%. Among them, device 4 has the lowest identification accuracy at 96%, indicating that the RF fingerprints of device 4 and device 14 are very similar. Additionally, there is a 1% misjudgment rate for device 5. Except for the above three devices, the identification accuracy for the rest devices can reach 100%. The above experiments demonstrate MRFE's exceptional performance in such challenging device identification scenarios.

G. Effect of FSL

FSL is the main component of the feature extraction module. It utilizes spatial weights and channel weights to enhance the fingerprints extracted by convolution kernels. To verify its enhancement on RF fingerprints, we conduct ablation experiments by replacing FSL with traditional convolutional layers, as shown in Fig. 9. In addition, we also verify the wide application of FSL on the AlexNet network. In the experiments, we use four FSLs containing 64, 128, 128, and 64 kernels, respectively. The four convolutional layers used to replace FSL have the same number of convolutional kernels.

We first verify the effect of FSL on each device in the MRFE. Experimental results in Fig. 9 demonstrate an accuracy

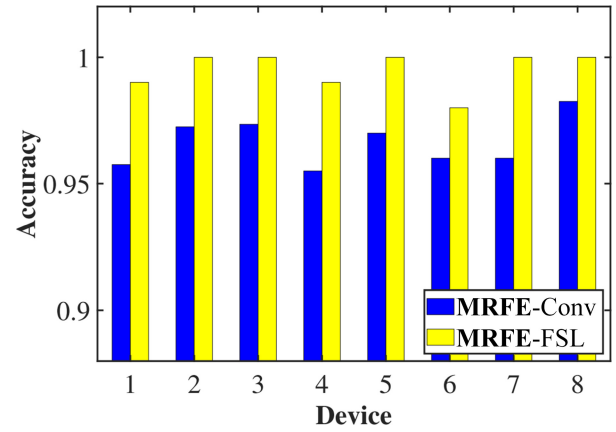


Fig. 9. Device identification accuracy of MRFE applying FSL and Conv.

TABLE IV
EXPERIMENTAL RESULTS OF ALEXNET AND MRFE
APPLICATION CONV AND FSL

Method	Model	Accuracy	Precision	Recall	F1 score
Conv	AlexNet	93.84%	93.79%	93.85%	93.82%
	ResNet	96.26%	96.38%	96.13%	96.25%
	Inception	97.03%	96.73%	97.32%	97.02%
	MRFE	97.28%	97.26%	97.29%	97.27%
FSL	AlexNet	95.63%	95.75%	95.62%	95.68%
	ResNet	97.82%	97.26%	97.95%	97.60%
	Inception	98.23%	98.48%	98.06%	98.26%
	MRFE	99.33%	99.32%	99.35%	99.33%

increase for each device with FSL usage. Particularly, device 4 and shows the most significant increase in accuracy, at 4%. Additionally, to validate FSL's effectiveness on other networks, its application in place of convolutional layers in the AlexNet network is tested. The experimental outcomes, detailed in Table IV, reveal an average accuracy improvement of 2%. These experiments confirm that FSL is a versatile network plugin capable of enhancing fingerprint representation.

H. Effect of Signal Branches

Three signal channels, i.e., the I signal channel, the Q signal channel, and the mixed-signal channel, are key components of MRFE. The I signal channel extracts the fingerprint from the hardware path generating the I signal. Similarly, the Q signal channel is tasked with extracting the fingerprint from the hardware path producing the Q signal. The mixed-signal channel extracts all fingerprints from the transmitter hardware. In this section, we conduct ablation experiments to analyze the roles of the I signal and Q signal channels.

In Fig. 10, MRFE-without IQ represents the MRFE variant lacking both the I and Q signal channels. MRFE-without I denotes a model variant without the I signal channel while MRFE-without Q refers to a model variant without the Q signal channel. It can be seen from Fig. 10 that after 20 epochs of training, the accuracy of MRFE-without IQ is the lowest among the four models, only reaching 86%, indicating that simply using mixed-signal channels cannot achieve the

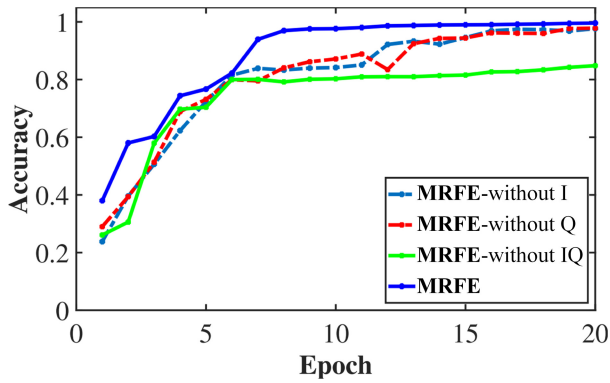


Fig. 10. Training process with different signal channels.

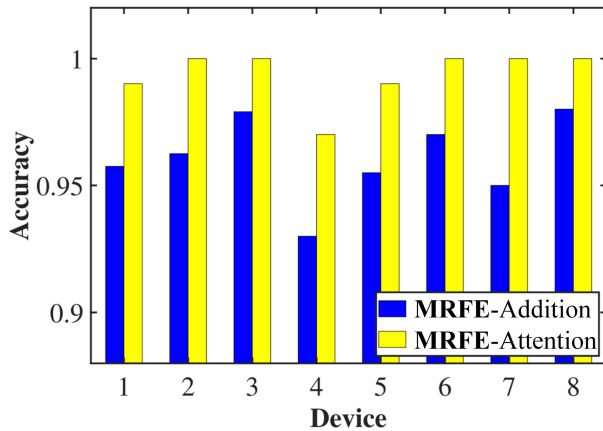


Fig. 11. Device identification accuracy of using addition and attention.

ideal accuracy. With the addition of the I signal channel or Q signal channel, the accuracy of the MRFE is significantly improved, which achieves 93%. Furthermore, by combining information from the three channels, the MRFE achieves both faster convergence rate and higher accuracy. The above experiments demonstrate that both the I signal channel and Q signal channel can enhance the identification capability of MRFE.

I. Effect of Attention

The attention mechanism is an essential component in the fingerprint fusion modules. It fuses three signal fingerprints extracted from the I signal channel, the Q signal channel, and the mixed-signal channel. The attention layer can enhance the mixed-signal fingerprint's representation by utilizing information from both I and Q signal fingerprints. In this section, we conduct comparative experiments to verify the enhancement effect of the attention mechanism on RF fingerprints.

In this experiment, we use addition instead of the attention mechanism to fuse the I signal fingerprint, Q signal fingerprint, and mixed-signal fingerprint. Experimental results, shown in Fig. 11, reveal that using the attention mechanism increases the accuracy for each device. Among them, the accuracy of device 7 increases the most, with an increase of 5%. The overall accuracy of device recognition has improved from 97.62%

to 99.33%. The above experimental results demonstrate that the attention mechanism can effectively fuse the three signal fingerprints and improve the wireless device recognition ability of MRFE.

VI. CONCLUSION

In this article, we present MRFE, a novel RF fingerprinting enhancement technique based on the hardware structure and inherent features within the transmitter chain for accurate IoT device identification. Extensive experimental results demonstrate that, in challenging identification scenarios involving a large pool of bit-similar devices, our MRFE achieves an impressive accuracy rate of 99.33% using only raw IQ samples at physical layer. The rapid recognition feature of MRFE is particularly beneficial in scenarios that require rapid response and cautious resource management, highlighting its outstanding scalability and adaptability in various IoT applications and services. In the future, we aim to integrate a denoising module into MRFE to reduce noise interference and incorporate demodulated symbols to enhance feature sets for more robust device identification under challenging conditions.

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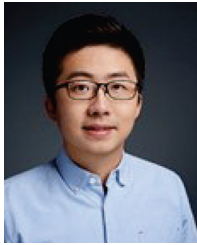
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